

1 Functions and Relations

Proposition 1. *If n is a positive integer, then 133 divides $11^{n+1} + 12^{2n-1}$.*

Proof. Note that $133 = 7 \cdot 19$ and $\gcd(7, 19) = 1$. It suffices to show

$$11^{n+1} + 12^{2n-1} \equiv 0 \pmod{7} \quad \text{and} \quad 11^{n+1} + 12^{2n-1} \equiv 0 \pmod{19}.$$

Modulo 7. We have $11 \equiv 4 \pmod{7}$ and $12 \equiv 5 \pmod{7}$. Thus

$$11^{n+1} + 12^{2n-1} \equiv 4^{n+1} + 5^{2n-1} \pmod{7}.$$

Since $5 \equiv -2$ and $25 \equiv 4 \pmod{7}$,

$$5^{2n-1} = 5(5^2)^{n-1} \equiv 5 \cdot 4^{n-1} \pmod{7}.$$

Hence

$$4^{n+1} + 5^{2n-1} \equiv 4^{n+1} + 5 \cdot 4^{n-1} = 4^{n-1}(4^2 + 5) = 4^{n-1}(16 + 5) = 4^{n-1} \cdot 21 \equiv 0 \pmod{7}.$$

Modulo 19. We have $12^2 = 144 \equiv 11 \pmod{19}$, so $11 \equiv 12^2 \pmod{19}$ and hence

$$11^{n+1} \equiv (12^2)^{n+1} = 12^{2n+2} \pmod{19}.$$

Therefore

$$11^{n+1} + 12^{2n-1} \equiv 12^{2n+2} + 12^{2n-1} = 12^{2n-1}(12^3 + 1) \pmod{19}.$$

Compute $12^3 \pmod{19}$:

$$12^2 \equiv 11, \quad 12^3 \equiv 11 \cdot 12 = 132 \equiv -1 \pmod{19}.$$

Thus $12^3 + 1 \equiv -1 + 1 \equiv 0 \pmod{19}$, and so

$$11^{n+1} + 12^{2n-1} \equiv 12^{2n-1} \cdot 0 \equiv 0 \pmod{19}.$$

We have shown that $11^{n+1} + 12^{2n-1}$ is divisible by both 7 and 19. Since $\gcd(7, 19) = 1$, it follows that $11^{n+1} + 12^{2n-1}$ is divisible by $7 \cdot 19 = 133$. Hence $133 \mid 11^{n+1} + 12^{2n-1}$ for every positive integer n . $\square$

Proposition 2. *Define $R \subseteq \mathbb{R} \times \mathbb{R}$ by $(a, b) \in R \iff a - b \in \mathbb{Q}$.*

Proof. (i) R is an equivalence relation.

Reflexive: $a - a = 0 \in \mathbb{Q} \Rightarrow (a, a) \in R$.

Symmetric: $a - b \in \mathbb{Q} \Rightarrow b - a = -(a - b) \in \mathbb{Q} \Rightarrow (b, a) \in R$.

Transitive: $a - b \in \mathbb{Q}, b - c \in \mathbb{Q} \Rightarrow a - c = (a - b) + (b - c) \in \mathbb{Q} \Rightarrow (a, c) \in R$.

Hence R is an equivalence relation.

(ii) Cardinality of equivalence classes.

The class of a is $[a] = a + \mathbb{Q} = \{a + q : q \in \mathbb{Q}\}$. Since $\mathbb{Q}$ is countable, each class $[a]$ is countable.

If the set of classes were countable, then $\mathbb{R}$, being the union of all classes, would be a countable union of countable sets and therefore countable — contradiction. Thus the set of equivalence classes is uncountable. $\square$

Proposition 3. *There do not exist positive integers x, y satisfying*

$$x^2 = 2y^2 + 3.$$

Proof. Reduce modulo 4. Squares are 0 or 1 (mod 4). If y were even then $2y^2 \equiv 0 \pmod{4}$ and hence $x^2 = 2y^2 + 3 \equiv 3 \pmod{4}$, impossible. Thus y is odd. Write $y = 2k + 1$. Then

$$y^2 = (2k + 1)^2 = 4k(k + 1) + 1 \equiv 1 \pmod{8}$$

because $k(k + 1)$ is even, so $4k(k + 1)$ is a multiple of 8. Hence

$$2y^2 + 3 \equiv 2 \cdot 1 + 3 = 5 \pmod{8}.$$

But no square is congruent to 5 (mod 8) (odd squares are 1 (mod 8), even squares are 0 or 4 (mod 8)). This contradiction shows there are no positive integers x, y with $x^2 = 2y^2 + 3$. $\square$

Proposition 4. *There are no positive integers x, y satisfying*

$$x^2 = 2y^2 + 3.$$

Proof. Work modulo 3. Squares mod 3 are 0 or 1.

Reduce the equation mod 3. Since $3 \equiv 0 \pmod{3}$ we obtain

$$x^2 \equiv 2y^2 \pmod{3}.$$

Case 1: $y \not\equiv 0 \pmod{3}$. Then $y^2 \equiv 1 \pmod{3}$, so $x^2 \equiv 2 \pmod{3}$, impossible because a square modulo 3 cannot be 2.

Case 2: $y \equiv 0 \pmod{3}$. Write $y = 3k$. Then

$$x^2 = 2(3k)^2 + 3 = 18k^2 + 3 = 3(6k^2 + 1),$$

so $3 \mid x^2$ and hence $3 \mid x$. Put $x = 3t$. Substituting and dividing by 3 yields

$$3t^2 = 6k^2 + 1 \implies 0 \equiv 1 \pmod{3},$$

a contradiction.

Both cases are impossible. Therefore no positive integers x, y satisfy $x^2 = 2y^2 + 3$. $\square$

Lemma 1. *For any integer n , the residue of $n^2 \pmod{3}$ is either 0 or 1.*

Proof. Every integer n is congruent modulo 3 to exactly one of

$$0, 1, 2.$$

Compute the squares of these representatives:

$$0^2 \equiv 0 \pmod{3}, \quad 1^2 \equiv 1 \pmod{3}, \quad 2^2 = 4 \equiv 1 \pmod{3}.$$

Thus every square is congruent to either 0 or 1 modulo 3. $\square$

Proposition 5. *For an integer n , the following are equivalent:*

- (i) n is odd.
- (ii) $n^2 \equiv 1 \pmod{8}$.
- (iii) $n^4 \equiv 1 \pmod{16}$.

Proof. **(i $\Rightarrow$ ii).** If n is odd, write $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k(k + 1) + 1.$$

Since $k(k + 1)$ is even, $4k(k + 1)$ is a multiple of 8, hence

$$n^2 \equiv 1 \pmod{8}.$$

(ii $\Rightarrow$ iii). Assume $n^2 \equiv 1 \pmod{8}$. Then $n^2 = 8m + 1$ for some $m \in \mathbb{Z}$. Square both sides:

$$n^4 = (8m + 1)^2 = 64m^2 + 16m + 1 \equiv 1 \pmod{16}.$$

(iii $\Rightarrow$ i). Assume $n^4 \equiv 1 \pmod{16}$. Reduce modulo 2:

$$n^4 \equiv 1 \pmod{2}.$$

(iii $\Rightarrow$ i). Assume $n^4 \equiv 1 \pmod{16}$. If n were even, write $n = 2k$; then

$$n^4 = (2k)^4 = 16k^4 \equiv 0 \pmod{16}.$$

Hence an even n cannot satisfy $n^4 \equiv 1 \pmod{16}$. Therefore n is odd.

Thus (i), (ii), (iii) are equivalent. □

Proposition 6. For all real numbers x, y we have

$$|x + y| \geq ||x| - |y||.$$

Proof. We split into two cases.

Case 1: $|x| \geq |y|$.

Then $|x| - |y| \geq 0$. Using the triangle inequality in the form $|x| = |(x + y) - y| \leq |x + y| + |y|$ we obtain

$$|x| - |y| \leq |x + y|.$$

Hence $||x| - |y|| = |x| - |y| \leq |x + y|$.

Case 2: $|x| < |y|$.

Then $|y| - |x| > 0$. Apply the same triangle-inequality trick with the roles of x and y interchanged: $|y| = |(x + y) - x| \leq |x + y| + |x|$, whence

$$|y| - |x| \leq |x + y|.$$

Thus $||x| - |y|| = |y| - |x| \leq |x + y|$.

In both cases $||x| - |y|| \leq |x + y|$. Therefore $|x + y| \geq ||x| - |y||$ for all real x, y . □

Proposition 7. Let $\mathcal{I} = \{A \subseteq \mathbb{N} : A \text{ is infinite}\}$ and define R on $\mathcal{I}$ by $A R B \iff A \subseteq B$.

1. R is a partial order on $\mathcal{I}$.
2. R is not a total (linear) order.
3. $\mathcal{I}$ has no minimum element, but it has a maximum element (namely $\mathbb{N}$).

Proof. (1) Reflexive: $A \subseteq A$ for every $A \in \mathcal{I}$. Antisymmetric: $A \subseteq B$ and $B \subseteq A$ imply $A = B$. Transitive: $A \subseteq B$ and $B \subseteq C$ imply $A \subseteq C$. Hence R is a partial order.

(2) Not total: take the infinite sets of even and odd naturals, $E = \{2, 4, 6, \dots\}$ and $O = \{1, 3, 5, \dots\}$. Then $E, O \in \mathcal{I}$ but neither $E \subseteq O$ nor $O \subseteq E$, so R is not linear.

(3) Minimum: suppose $M \in \mathcal{I}$ were least, i.e. $M \subseteq A$ for every infinite A . Choose any $m \in M$. The set $A = \mathbb{N} \setminus \{m\}$ is infinite, but $m \notin A$, hence $M \not\subseteq A$, contradiction. Thus no minimum exists.

Maximum: $\mathbb{N} \in \mathcal{I}$ and for every $A \in \mathcal{I}$ we have $A \subseteq \mathbb{N}$, so $\mathbb{N}$ is the greatest element (maximum). □

2 Group and Graph

Proposition 8. *Let G be a cyclic group of order 24. Then G has exactly 2 elements of order 4.*

Proof. Since G is cyclic of order 24, let $G = \langle g \rangle$. Every element of G is of the form g^k for $0 \leq k \leq 23$, and its order is

$$\text{ord}(g^k) = \frac{24}{\gcd(24, k)}.$$

We seek all k such that $\text{ord}(g^k) = 4$. Thus:

$$\frac{24}{\gcd(24, k)} = 4 \implies \gcd(24, k) = 6.$$

Now list all k with $0 \leq k < 24$ and $\gcd(24, k) = 6$. The multiples of 6 below 24 are 0, 6, 12, 18. Their gcds with 24 are:

$$\gcd(24, 0) = 24, \quad \gcd(24, 6) = 6, \quad \gcd(24, 12) = 12, \quad \gcd(24, 18) = 6.$$

Thus exactly two values satisfy $\gcd(24, k) = 6$, namely $k = 6$ and $k = 18$.

Therefore the only elements of order 4 are g^6 and g^{18} , and hence there are exactly 2 elements of order 4 in G . $\square$

Proposition 9. *The statement “Every connected bipartite graph has at least one vertex of even degree” is false.*

Proof. Consider the complete bipartite graph $K_{3,3}$. It is connected and bipartite with bipartition $\{a_1, a_2, a_3\} \cup \{b_1, b_2, b_3\}$, and $\deg(a_i) = \deg(b_j) = 3$ for all i, j . Thus every vertex has odd degree and there is no vertex of even degree. Hence the given statement is false. $\square$

Proposition 10. *Let G be a connected graph in which every block (i.e. maximal 2-connected subgraph) is bipartite. Then G is bipartite.*

Proof. Recall two standard facts:

- (1) A cycle is 2-connected, hence is contained in some block of G .
- (2) A graph is bipartite $\iff$ it contains no cycle of odd length.

Assume, for contradiction, that G is not bipartite. By (2) G then contains an odd cycle C . By (1) the cycle C is contained in some block B of G . Thus B contains an odd cycle, so B is not bipartite, which contradicts the hypothesis that every block of G is bipartite.

Hence G has no odd cycle. Again by (2), G is bipartite. $\square$